

Entry Header

Name:

Aswina Sivaloganathan

Short description:

Using GenAI to brainstorm ways to represent anxiety in a p5.js blurb, then modifying an existing sketch by applying subtle colour jitter to create visual instability.

Small decision or change made:

I chose to represent anxiety using subtle colour jitter rather than dramatic movement, refining the randomness so the effect felt tense instead of chaotic.

GenAI Documentation

GenAI tools were used for this task.

Date Used:

January 25th, 2026

Tool Disclosure:

ChatGPT 5.2

Appendix (Evidence)

Code Comment:

```
// Added subtle colour jitter to the fill values to create visual instability and represent anxiety without dramatic movement.
```

```
fill(20 + random(-30, 30), 120 + random(-30, 30), 255);
```

Transcript

For my sidequest, we need to pick an emotion and represent that on a "blurb" (its coded with p5), could you help me find ways to effectively demonstrate different emotions through that, such as anxiety, fear, or happiness. Give me a few different ideas

Yes — this is a **great SideQuest prompt** and p5.js is *perfect* for emotional expression. I'll give you **clear, code-translatable ideas** you can actually implement (not vague "vibes"). Below are **3 emotions (Anxiety, Fear, Happiness)** with **multiple distinct visual + interaction concepts** for each. You can pick **one emotion** and one concept, or even let the blurb *shift* emotions over time.

ANXIETY

Anxiety is about **instability, anticipation, overload, and loss of control.**

Idea 1: Shaking Grid / Unstable System

Visual

- A grid or repeated pattern (circles, lines, squares)
- Each element **jitters slightly**, but not in sync
- Movement increases the longer the sketch runs

Interaction

- Mouse movement increases jitter intensity
- If the mouse stops → jitter *continues anyway*

Why it works

- Suggests *trying to maintain order* but failing
- Feels restless and never fully settles

p5 tools

`noise()`

`random()`

`small frameCount-based offsets`

Idea 2: Breathing That Never Quite Calms

Visual

- A single shape (circle or blob) that expands and contracts like breathing
- But the rhythm is **irregular**
- Sometimes it expands too fast or pauses too long

Interaction

- Clicking tries to “calm” it
- Works temporarily → then anxiety creeps back

Why it works

- Mimics anxious breathing
 - False sense of control
-

Idea 3: Overstimulated Screen

Visual

- Multiple translucent shapes layered on top of each other
- Subtle flicker in opacity
- Text fragments or symbols drifting by (optional)

Interaction

- Moving mouse causes *more* elements to appear
- There’s no way to reduce clutter

Why it works

- Represents mental overload
 - No clear focal point
-



Fear is about **threat, uncertainty, darkness, and anticipation.**

Idea 1: Limited Visibility (Fog of Fear)

Visual

- Entire canvas is dark
- Only a small circular area around the mouse is visible
- Objects appear *just* at the edge of visibility

Interaction

- Moving the mouse reveals things
- Some shapes disappear when you look directly at them

Why it works

- Fear of the unknown
- Threat is implied, not shown

p5 tools

`blendMode()`

`masking via alpha`

`ellipse with alpha overlay`

Idea 2: Approaching Threat

Visual

- Distant shapes slowly move toward the center
- They're blurry or indistinct
- As they get closer, screen subtly darkens

Interaction

- Clicking pauses movement briefly
- Threat resumes regardless

Why it works

- Slow inevitability
 - Builds dread rather than jump scares
-

Idea 3: Sudden Disruptions

Visual

- Calm, minimal scene
- Occasionally, abrupt flashes, distortions, or color inversions

Interaction

- These disruptions are **random**, not user-triggered

Why it works

- Unpredictability = fear
 - User cannot prepare
-

HAPPINESS

Happiness is about **flow, harmony, warmth, responsiveness, and ease.**

Idea 1: Playful Motion System

Visual

- Soft shapes floating upward
- Bright but gentle colors
- Motion follows smooth curves

Interaction

- Mouse movement nudges shapes
- They respond immediately and positively

Why it works

- Feels light, playful, and rewarding
 - Clear cause and effect
-

Idea 2: Growing Garden

Visual

- Minimal scene at first
- Clicking “plants” shapes
- Over time they grow, branch, or multiply

Interaction

- User actions lead to visible, lasting growth

Why it works

- Happiness through creation and care
 - No penalties or failure
-

Idea 3: Rhythmic Celebration

Visual

- Shapes pulse in sync with a steady rhythm
- Color transitions are smooth and warm

Interaction

- Mouse or key presses add *new rhythms* instead of breaking old ones

Why it works

- Order + variety
- Encourages exploration

✨ Want an A+ Twist? (Highly Recommend)

Emotion that evolves over time

Example:

- Starts as **anxiety** (jittery, unstable)
- If user interacts calmly → transitions into **happiness**
- If ignored → intensifies into **fear**

This shows:

- Emotional systems
- Interaction design
- Narrative thinking

If you want, next I can:

- Help you **pick ONE emotion + concept** that's easiest to code
- Sketch a **p5.js structure** (without writing the full code for you)
- Help you write the **blurb text**

- Make sure it clearly satisfies SideQuest criteria

Just tell me which emotion you're leaning toward 🧐

I want to do anxiety. I would like to make minimal changes and do a jitter or a shake to represent anxious feelings. What kind of parameters in p5 might accurately demonstrate these movements, such as (fill of the blurb or noise).

This is a **strong, smart choice** — minimal jitter is *very* effective for anxiety, especially in a blurb where subtlety matters more than spectacle.

Below are **p5-specific parameters and motion ideas** you can think in terms of *conceptually*

Core Principle for Anxiety

Nothing is visibly “wrong,” but nothing is fully stable.

The movement should feel **involuntary**, **persistent**, and **slightly unpredictable**.

1. Position Jitter (Micro-Instability)

Use **very small positional offsets** that never fully stop.

Parameters to think about

- **Amplitude:** extremely small (1–4 pixels)
- **Frequency:** constant, but uneven
- **Direction:** slightly different on x vs y

Conceptual effect

- The blurb *tries* to sit still
- It never quite does

Why this reads as anxiety

- Mimics nervous energy
 - Feels subconscious, not expressive
-

2. Noise vs Random (Very Important Choice)

Random → panic / chaos

Noise → anxiety / tension

For anxiety:

- Use **slow noise-based drift**, not harsh randomness
- Noise changes gradually but never repeats cleanly

Parameters

- **Noise scale**: very small (slow change)
- **Time offset**: always advancing
- **Noise intensity**: capped (never extreme)

Conceptual metaphor

- Thoughts looping
 - Restlessness without resolution
-

3. Subtle Rotation Shake

Instead of just moving position, add **tiny rotation shifts**.

Parameters

- **Rotation range**: $\pm 0.5^\circ$ to $\pm 2^\circ$
- **Speed**: irregular, slightly faster than breathing

- **Center:** rotates around its own center (not the canvas)

Why this works

- Rotation feels *physical*
 - Reads like internal tension instead of external motion
-

4. Fill / Opacity Instability

This is HUGE for anxiety and very minimal visually.

Fill ideas

- Slight opacity flicker
- Tiny shifts in brightness or saturation
- Color stays “safe” but unstable

Parameters

- **Opacity variance:** 5–15%
- **Timing:** asynchronous with movement
- **Never fully opaque or transparent**

Emotional read

- Feels like a heartbeat or breath that’s off rhythm
 - Suggests hyper-awareness
-

5. Rhythm That Refuses to Settle

Avoid smooth sine waves.

Instead:

- Use **almost rhythmic motion**
- Occasionally interrupt or speed up briefly
- Then return to baseline

Parameters

- **Base rhythm:** slow
- **Interruptions:** brief, subtle
- **Recovery:** incomplete (never full calm)

Why this matters

- Anxiety isn't chaos — it's *anticipation*
-

6. Accumulating Tension Over Time

This is powerful and easy conceptually.

Idea

- Jitter **slowly increases** the longer the sketch runs
- User does nothing → anxiety builds

Parameters

- **Time-based scaling**
- **Upper limit** (so it doesn't become panic)

Conceptual meaning

- Unresolved anxiety compounds
- Still minimal, but noticeable

7. Mouse Interaction (Optional, Very Minimal)

If you include interaction, keep it subtle.

Good interaction

- Mouse proximity slightly increases jitter
- Clicking attempts to stabilize → works briefly

Avoid

- Full control
- Clear “fix”

Why

- Anxiety is not easily resolved
- Temporary relief reads as realistic

8. What NOT to Do (Important)

Avoid:

- Large movement
- Sudden jumps
- Strong color contrast
- Perfect loops

These shift the emotion toward:

- Panic
- Fear
- Excitement

If I wanted to implement subtle colour jitter in p5, how could I apply small randomness to the RGB values in `fill()` so the colour feels unstable but still cohesive? Am I able to do this with just only the function `fill()`?

Yes — **you can absolutely do this using just `fill()`**, and subtle colour jitter is a *very good* choice for anxiety because it reads as **instability without motion**.

Below are **clear, safe patterns** you can use, from simplest → most refined. All of these keep the colour **cohesive**, not chaotic.

Core Rule for Colour Anxiety

Small range, same base colour, continuous variation.

If the colour jumps too far, it becomes excitement or glitch — not anxiety.

1 Simple RGB Micro-Jitter (Most Direct)

This is the most straightforward and totally valid for a SideQuest.

```
fill(  
  20 + random(-5, 5),  
  120 + random(-5, 5),  
  255 + random(-5, 5)  
);
```

Why this works

- The **base colour stays the same**
- The variation is **tiny**
- The jitter feels involuntary

How to tune it

- $\pm 2-4$ → very subtle tension
- $\pm 5-8$ → noticeable unease
- $\pm 10+$ → starts to feel glitchy

For anxiety, **stay under ± 6 .**

2 Noise-Based Colour Instability (More Natural)

If you want it to feel *alive* rather than flickery.

```
let n = noise(frameCount * 0.01);
```

```
fill(  
  20 + n * 6,  
  120 + n * 6,  
  255 + n * 6  
);
```

Why this is better for anxiety

- Noise changes **slowly**
- No harsh flicker
- Feels like tension building under the surface

Design meaning

This mimics:

- fluctuating focus
 - background nervous energy
 - emotional “buzz”
-

③ Asymmetric Channel Jitter (Very Effective)

Instead of jittering all RGB equally, offset them slightly.

```
fill(  
    20 + random(-3, 3),  
    120 + random(-5, 5),  
    255 + random(-2, 2)  
);
```

Why this reads as anxiety

- Colour *almost* stays the same
- Something feels “off”
- Subconscious discomfort

This is excellent for **minimal designs**.

④ Opacity Jitter (Extremely Subtle, Very Clean)

If you want almost invisible anxiety.

```
fill(20, 120, 255, 180 + random(-10, 10));
```

Why this is powerful

- No colour change
- Only presence feels unstable
- Reads as internal tension

This works especially well on text or blurb shapes.

5 Controlled Jitter with **constrain()** (Safety Net)

If you're worried about colours drifting too far:

```
fill(  
    constrain(20 + random(-5, 5), 15, 25),  
    constrain(120 + random(-5, 5), 115, 125),  
    constrain(255 + random(-5, 5), 245, 255)  
);
```

Why graders like this

- Shows **intentional control**
 - Prevents visual chaos
 - Signals design thinking
-

What NOT to Do (Important)

Avoid:

```
fill(random(255), random(255), random(255));
```

This reads as:

- excitement
 - chaos
 - glitch art
 - ✗ not anxiety
-

Best Combo for Anxiety (If You Pick One)

If you want the **cleanest, most defensible approach**:

- **Noise-based RGB jitter**
 - **Small range ($\pm 4-6$)**
 - **Optional opacity fluctuation**
-

If you want, next I can:

- Help you **pick exact numbers** so it reads right
- Check whether your current colour choice supports anxiety
- Help you write the **1-2 sentence SideQuest reflection**
- Help you decide between **random vs noise** for grading clarity

Just tell me what your base colour is 🎨